

BS-CH101 / BS-CH201 — Chemistry-I

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Module 6 Exam Notes: Stereochemistry

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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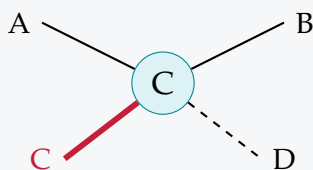
1. Stereochemistry

Stereochemistry is the study of three-dimensional arrangement of atoms in molecules and how this arrangement affects physical, chemical, and biological properties.

1.1 Representations of Three-Dimensional Structures

Three-dimensional structures are represented using wedge-dash, Fischer projection, Newman projection, and sawhorse projection.

Wedge-Dash Representation Around a Tetrahedral Carbon



solid wedge: coming out; dashed bond: going back

1.2 Structural Isomers and Stereoisomers

Structural isomers differ in connectivity, whereas stereoisomers have same connectivity but different spatial arrangement.

Type	Meaning	Example
Chain isomerism	Different carbon skeleton	n-butane and isobutane
Position isomerism	Functional group at different position	1-propanol and 2-propanol
Geometrical isomerism	Different arrangement around double bond/ring	cis-2-butene and trans-2-butene
Optical isomerism	Non-superimposable mirror images	lactic acid enantiomers

1.3 Chirality, Enantiomers and Diastereomers

A molecule is chiral if it is not superimposable on its mirror image. A common cause is a carbon attached to four different groups.

Enantiomers are non-superimposable mirror-image stereoisomers. **Diastereomers** are stereoisomers that are not mirror images.

1.4 Optical Activity and Absolute Configuration

Optically active substances rotate plane-polarized light. Absolute configuration is assigned by Cahn-Ingold-Prelog rules.

Specific Rotation

$$[\alpha]_{\lambda}^T = \frac{\alpha}{lc}$$

where α is observed rotation in degrees, l is path length in dm, and c is concentration in g mL^{-1} .

1.5 Elements of Symmetry

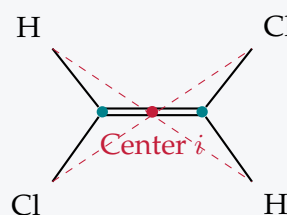
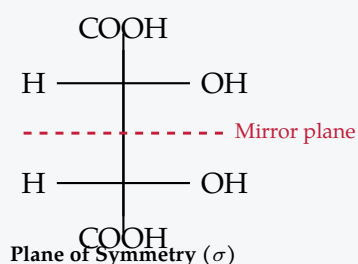
To understand chirality and optical activity, we must analyze the symmetry elements of a molecule. A molecule is chiral (and thus optically active) if and only if it lacks an **alternating axis of symmetry** (S_n). In practice, for most organic molecules, this is equivalent to lacking a **plane of symmetry** (σ) and a **center of symmetry** (i).

Symmetry Elements

The three primary elements of symmetry are:

- **Plane of Symmetry** (σ): An imaginary plane that bisects a molecule into two halves that are mirror images of each other. Examples include meso-tartaric acid and chlorobenzene.
- **Center of Symmetry** (i): A point within a molecule such that any line drawn from any atom through this point meets an identical atom at an equal distance on the opposite side. Examples include trans-1,2-dichloroethylene and trans-1,4-dimethylcyclohexane.
- **Alternating Axis of Symmetry** (S_n): An axis such that rotation of the molecule by $360^\circ/n$ followed by reflection across a plane perpendicular to the axis results in a configuration identical to the original.

Symmetry Elements in Common Molecules



Chirality and Symmetry Rule

A molecule is chiral and optically active if and only if it is **dissymmetric** (lacks an alternating axis of symmetry, S_n).

$$\text{Chirality} \iff \text{Absence of } S_n \quad (\text{Practically: absence of } \sigma \text{ and } i)$$

Note: A molecule can have a simple axis of symmetry (C_n) and still be chiral. For example, trans-1,2-dimethylcyclopropane has a C_2 axis but no σ or i , making it chiral and optically active.

1.6 Conformational Analysis

Conformations arise by rotation about single bonds. Ethane has staggered and eclipsed conformations; staggered is more stable due to lower torsional strain.

Ethane Conformations: Staggered and Eclipsed

Staggered



Eclipsed



1.7 Isomerism in Transition Metal Compounds

Coordination compounds show geometrical, optical, linkage, ionization, and coordination isomerism. Square planar and octahedral complexes commonly show cis-trans isomerism.

Exam Points

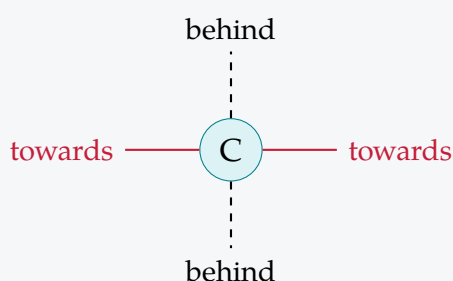
- Always distinguish structural isomerism from stereoisomerism.
- In optical isomerism, mention chirality and non-superimposable mirror images.
- CIP assignment needs priority order before clockwise/anticlockwise decision.
- For conformational analysis, compare stability using torsional strain.

2. Detailed Stereochemical Analysis

2.1 Fischer Projection Rules

Fischer projections are used mainly for molecules with chiral carbon chains. Horizontal bonds come out of the plane toward the viewer; vertical bonds go behind the plane.

Fischer Projection Convention



Rules for Fischer Projections

- Rotation by 180° gives the same molecule.
- Rotation by 90° gives a different configuration and is not allowed.

- Interchanging any two groups once inverts configuration.
- Interchanging two pairs of groups gives the original configuration.

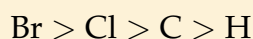
2.2 Assigning R and S Configuration

Cahn-Ingold-Prelog priority rules:

1. Atom directly attached with higher atomic number gets higher priority.
2. If first atoms are same, compare the next set of attached atoms.
3. Multiple bonds are treated as equivalent single bonds to duplicate atoms.
4. View the molecule with the lowest priority group pointing away.
5. Clockwise order $1 \rightarrow 2 \rightarrow 3$ is *R*; anticlockwise is *S*.

Worked Method: R/S Assignment

For a chiral carbon attached to Br, Cl, CH₃, and H:



Place H away from the viewer. If the order $\text{Br} \rightarrow \text{Cl} \rightarrow \text{CH}_3$ is clockwise, the configuration is *R*; if anticlockwise, it is *S*.

2.3 Geometrical Isomerism: E/Z and Cis/Trans

Geometrical isomerism occurs due to restricted rotation around a double bond or within rings. Cis/trans notation is used for simple cases; E/Z notation is used when each double-bond carbon has two different groups.

Notation	Meaning	Use case
Cis	Similar groups on same side	Simple alkenes or rings
Trans	Similar groups on opposite sides	Simple alkenes or rings
E	Higher-priority groups on opposite sides	General alkenes by CIP rule
Z	Higher-priority groups on same side	General alkenes by CIP rule

2.4 Meso Compounds

A meso compound contains chiral centres but is optically inactive due to internal symmetry. It is superimposable on its mirror image because optical rotations of two halves cancel internally.

Meso compound: An achiral molecule with two or more stereocentres and an internal plane of symmetry.

2.5 Conformational Analysis of Butane

Butane has anti, gauche, and eclipsed conformations. Anti is most stable because bulky methyl groups are far apart.

Conformation	Dihedral angle between methyl groups	Relative stability
Anti staggered	180°	Most stable
Gauche staggered	60°	Less stable due to steric repulsion
Eclipsed	0° or 120°	Least stable due to torsional strain

Specific Rotation Numerical

If observed rotation $\alpha = +6^\circ$, path length $l = 2 \text{ dm}$, and concentration $c = 0.5 \text{ g mL}^{-1}$:

$$[\alpha] = \frac{\alpha}{lc} = \frac{6}{2 \times 0.5} = +6^\circ$$

2.6 Coordination Compound Stereoisomerism

Square planar complexes of type MA_2B_2 show cis-trans isomerism. Octahedral complexes may show cis-trans, fac-mer, and optical isomerism depending on ligand arrangement.

Complex type	Possible isomerism	Example pattern
Square planar MA_2B_2	Cis-trans	$[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$
Octahedral MA_4B_2	Cis-trans	Two B ligands adjacent or opposite
Octahedral MA_3B_3	fac-mer	Three same ligands on one face or meridian

Stereochemistry Answer Checklist

1. Identify whether connectivity changes or only spatial arrangement changes.
2. Check for chirality, plane of symmetry, and mirror-image relation.
3. Use CIP rules for R/S or E/Z configuration.
4. Mention optical activity only after checking symmetry.

2.7 Number of Stereoisomers

For a molecule with n independent chiral centres, the maximum number of stereoisomers is:

$$2^n$$

If the molecule has internal symmetry, the actual number is less due to meso forms.

Solved Example: Number of Stereoisomers

A molecule has two chiral centres and no plane of symmetry.

$$\text{Maximum stereoisomers} = 2^2 = 4$$

These are arranged as two pairs of enantiomers. If a plane of symmetry exists, one or more meso forms reduce the total count.

2.8 Resolution of Racemic Mixtures

A racemic mixture contains equal amounts of two enantiomers and is optically inactive. Resolution means separation of enantiomers.

Method	Principle	Use
Mechanical separation	Physical separation of crystals	Rare; possible if crystals differ visibly
Chemical resolution	Formation of diastereomeric salts	Common laboratory method
Biochemical resolution	Enzyme reacts with one enantiomer faster	High selectivity
Chromatographic resolution	Chiral stationary phase separates enantiomers	Analytical and preparative use

2.9 Optical Purity and Enantiomeric Excess

Optical purity compares observed rotation with rotation of the pure enantiomer.

Optical Purity

$$\text{Optical purity} = \frac{\alpha_{\text{observed}}}{\alpha_{\text{pure}}} \times 100\%$$

For a two-enantiomer mixture, optical purity is equal to enantiomeric excess.

Solved Example: Enantiomeric Excess

If pure enantiomer has specific rotation $+40^\circ$ and mixture shows $+20^\circ$:

$$\text{optical purity} = \frac{20}{40} \times 100 = 50\%$$

So enantiomeric excess is 50%. The mixture contains 75% major enantiomer and 25% minor enantiomer.

2.10 Sequence Rules for Double Bonds: E/Z Worked Method

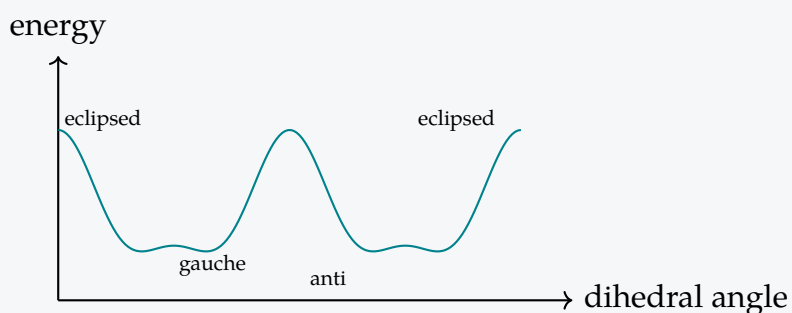
For each double-bond carbon, assign priority to its two substituents using CIP rules. If the two higher-priority groups are on the same side, the alkene is *Z*; if opposite, it is *E*.

Worked Method: E/Z Assignment

For $\text{CH}_3\text{CH}=\text{CHCl}$, compare substituents on each double-bond carbon. On the left carbon, CH_3 has higher priority than H. On the right carbon, Cl has higher priority than H. If CH_3 and Cl are on opposite sides, the configuration is *E*; if same side, it is *Z*.

2.11 Energy Profile of Butane Rotation

Rotation about the central C–C bond of butane gives different conformations. Anti is most stable; fully eclipsed is least stable.

Qualitative Energy Profile for Butane Rotation**2.12 Stereochemistry in Biological and Drug Action**

Many drug molecules are chiral, and different enantiomers can show different biological activity because enzymes and receptors are chiral environments.

Point	Importance
Different receptor fit	One enantiomer may bind better than the other
Different metabolism	Enzymes may break down enantiomers at different rates
Different toxicity	One enantiomer may be useful while the other causes side effects
Quality control	Optical purity matters in pharmaceutical synthesis

Previous-Year Style Questions

1. Explain chirality, enantiomers and diastereomers with examples.
2. State CIP rules and assign R/S configuration for a given molecule.
3. Explain optical activity, specific rotation and racemic mixture.
4. Compare conformations of ethane and butane.
5. Discuss geometrical and optical isomerism in coordination compounds.

Quick Revision — Module 6

Stereochemistry | BS-CH101

Key Formulas at a Glance

Formula / Rule	Meaning
$[\alpha]_{\lambda}^T = \frac{\alpha}{lc}$	Specific rotation of optically active compound.
$\text{enantiomeric excess} = \frac{ R - S }{R + S} \times 100\%$	Optical purity of a mixture.
$\Delta E = E_{\text{eclipsed}} - E_{\text{staggered}}$	Torsional energy barrier in conformational analysis.
CIP priority	Higher atomic number gets higher priority.

One-Line Definitions

- **Stereoisomers:** Same connectivity but different spatial arrangement.
- **Chiral molecule:** Molecule not superimposable on its mirror image.
- **Enantiomers:** Non-superimposable mirror-image isomers.
- **Diastereomers:** Stereoisomers that are not mirror images.
- **Optical activity:** Ability to rotate plane-polarized light.
- **Conformation:** Spatial arrangement formed by rotation about single bond.
- **Racemic mixture:** Equimolar mixture of two enantiomers.

Frequently Asked / Viva Questions

Q1. What is stereochemistry?

Study of three-dimensional arrangement of atoms in molecules.

Q2. What is chirality?

Non-superimposability of a molecule on its mirror image.

Q3. Define enantiomers.

Enantiomers are non-superimposable mirror-image stereoisomers.

Q4. Define diastereomers.

Diastereomers are stereoisomers that are not mirror images.

Q5. What is optical activity?

Rotation of plane-polarized light by a chiral substance.

Q6. What is a racemic mixture?

Frequently Asked / Viva Questions (contd.)

A 1:1 mixture of enantiomers having zero net optical rotation.

Q7. What does R/S configuration indicate?

It gives absolute three-dimensional arrangement around a chiral centre.

Q8. Why is staggered ethane more stable than eclipsed ethane?

Staggered conformation has minimum torsional repulsion.

Q9. Name isomerism in coordination compounds.

Geometrical, optical, linkage, ionization, and coordination isomerism.

Q10. What is geometrical isomerism?

Isomerism due to restricted rotation causing cis-trans or fac-mer forms.

Q11. What are the elements of symmetry that determine chirality?

A plane of symmetry (σ), center of symmetry (i), and alternating axis of symmetry (S_n). A molecule is chiral if and only if it lacks any S_n axis.

Q12. Can a molecule containing chiral carbon atoms be optically inactive?

Yes, meso compounds contain chiral carbons but are optically inactive because they have an internal plane of symmetry (σ), which causes internal compensation of optical rotation.

Summary Reference Table

Pair	Relation	Key difference
Structural isomers	Different connectivity	Different bonding sequence
Enantiomers	Mirror images	Equal and opposite optical rotation
Diastereomers	Not mirror images	Different physical properties
Conformers	Rotation about single bond	Interconvert without bond breaking